

Email Traffic Forensics

SMTP

Traffic files: [smtp.pcap](#)



Overview

- Simple Mail Transfer Protocol
- Hands-on lab
 - A .pcap file contains MS Outlook email traffic
 - Assume the traffic is decreased

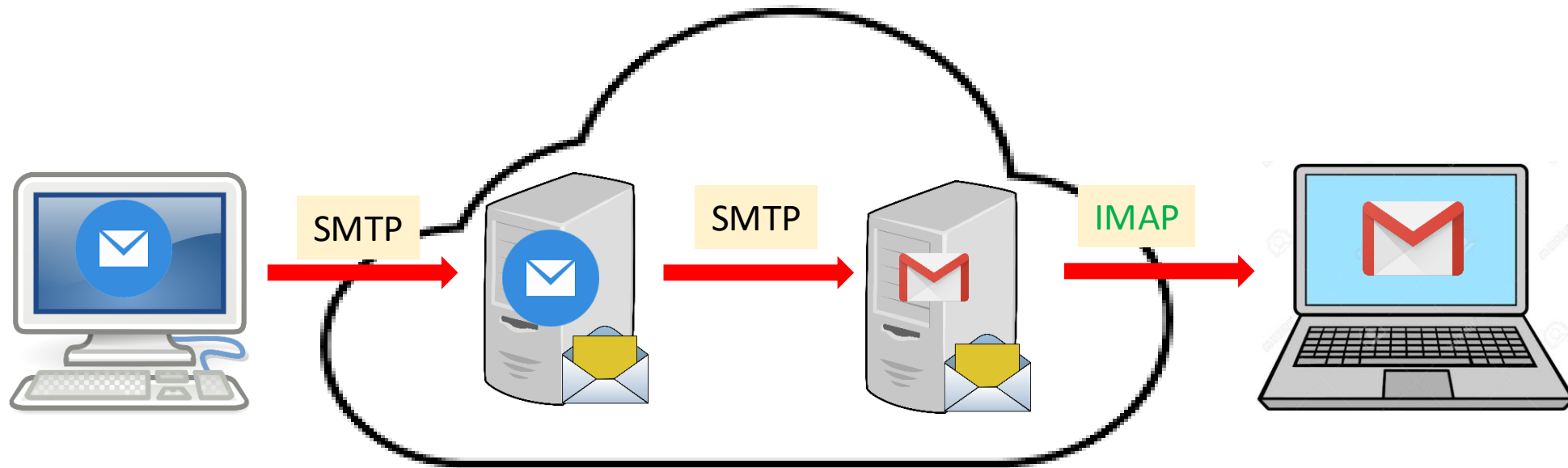


Simple Mail Transfer Protocol



Simple Mail Transfer Protocol (SMTP)

- It is a protocol used for sending and receiving email messages over the Internet.



SMTP (Simple Mail Transfer Protocol): sending emails

IMAP (Internet Access Message Protocol): **download** emails

Simple Mail Transfer Protocol

- SMTP operates on port **25** by default, but other ports such as **587** and **465** may also be used.
- SMTP uses a set of commands and responses to transfer email messages between clients and servers.
- SMTP is a plain text protocol
 - data transmitted between clients and servers is not encrypted by default.
- SMTPS for secure email transmissions
 - SMTP over SSL/TLS
 - STARTTLS

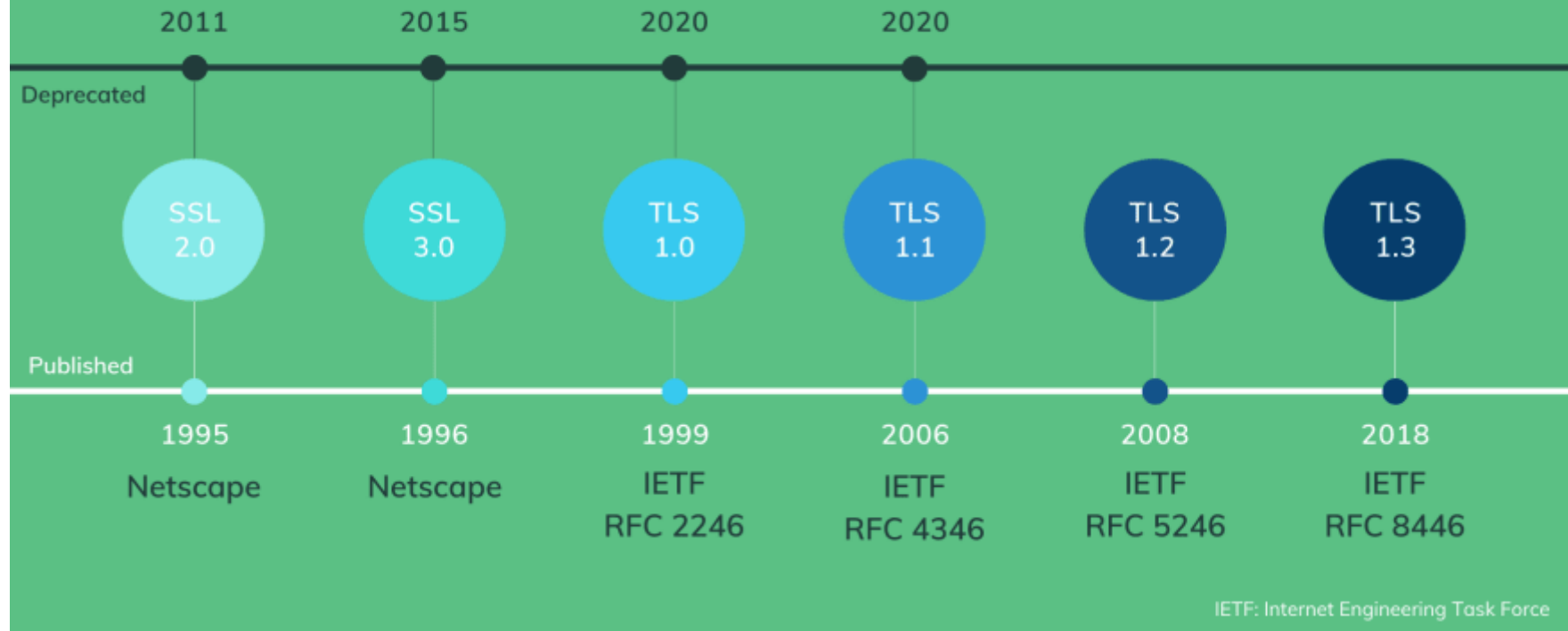


SSL/TLS

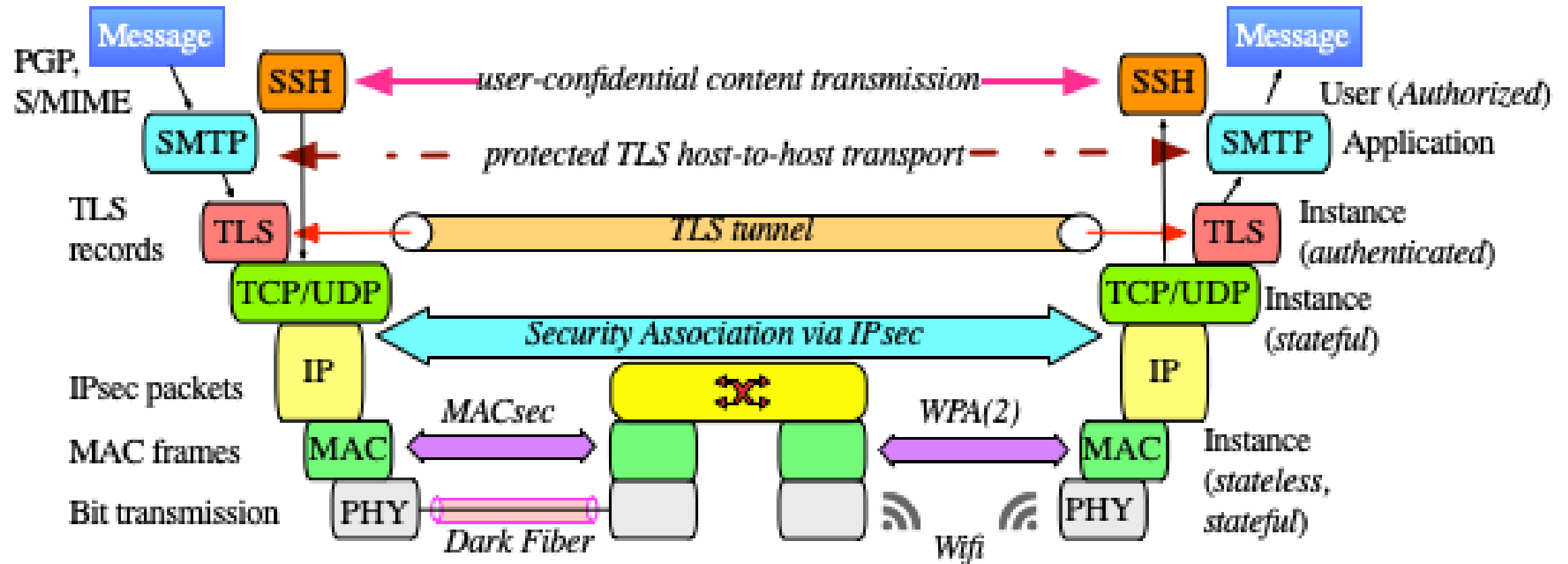
- SSL (Secure Sockets Layer) and TLS (Transport Layer Security) are cryptographic protocols that provide secure communication over the internet.
 - SSL is the predecessor to TLS, and is no longer considered secure.
- Key security features
 - Authentication: SSL and TLS provide mechanisms for authenticating the identity of the client and server, so that both parties can be sure they are communicating with the intended party.
 - Data encryption: SSL and TLS encrypt data transmitted over the internet, so that it cannot be intercepted and read by unauthorized parties.
 - Data integrity: SSL and TLS ensure the integrity of data transmitted over the internet, so that it cannot be modified in transit without detection.



THE HISTORY OF SSL / TLS



SMTP is an application layer protocol



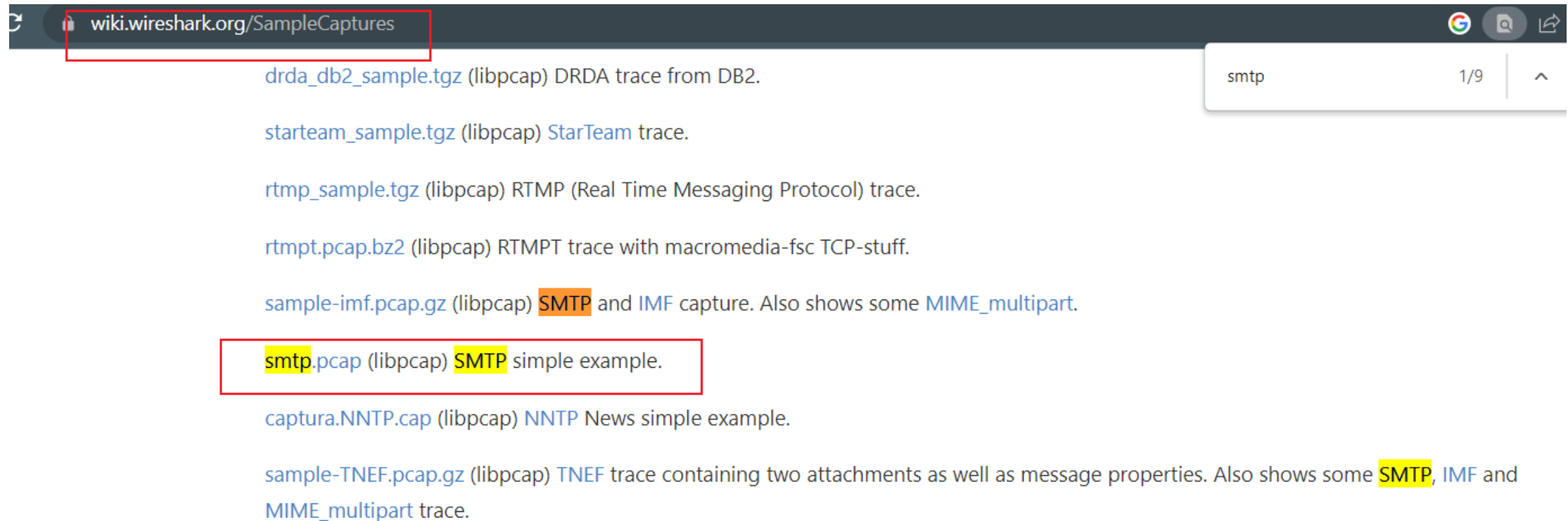
Hands-on Lab



Capture Emails traffic With Wireshark



Lab Files



<https://wiki.wireshark.org/SampleCaptures>



Download smtp.pcap

```
(kali㉿kali)-[~/smtp]
└─$ wget https://wiki.wireshark.org/uploads/__moin_import__/attachments/SampleCaptures/smtp.pcap
--2023-03-10 14:58:08-- https://wiki.wireshark.org/uploads/__moin_import__/attachments/SampleCaptures/smtp.pcap
Resolving wiki.wireshark.org (wiki.wireshark.org)... 172.67.75.39, 104.26.10.240, 104.26.11.240, ...
Connecting to wiki.wireshark.org (wiki.wireshark.org)|172.67.75.39|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 27850 (27K) [application/octet-stream]
Saving to: 'smtp.pcap'

smtp.pcap          100%[=====>] 27.20K  --.-KB/s    in 0.004s

2023-03-10 14:58:08 (6.91 MB/s) - 'smtp.pcap' saved [27850/27850]
```



SMTP first response “service ready”

3	0.036986	10.10.1.4	74.53.140.153	TCP	62	1470 → 25 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 SACK_PERM=1
4	0.383936	74.53.140.153	10.10.1.4	TCP	62	25 → 1470 [SYN, ACK] Seq=0 Ack=1 Win=5840 Len=0 MSS=1460 SACK_PERM=1
5	0.383968	10.10.1.4	74.53.140.153	TCP	54	1470 → 25 [ACK] Seq=1 Ack=1 Win=65535 Len=0
6	0.727603	74.53.140.153	10.10.1.4	SMTP	235	S: 220-xc90.websitewelcome.com ESMTP Exim 4.69 #1 Mon, 05 Oct 2009 01:05:54 -0500 \r\n
7	0.732749	10.10.1.4	74.53.140.153	SMTP	63	C: EHLO GP
8	1.073326	74.53.140.153	10.10.1.4	TCP	60	25 → 1470 [ACK] Seq=182 Ack=10 Win=5840 Len=0
9	1.074123	74.53.140.153	10.10.1.4	SMTP	191	S: 250-xc90.websitewelcome.com Hello GP [122.162.143.157] SIZE 5242
10	1.076669	10.10.1.4	74.53.140.153	SMTP	66	C: AUTH LOGIN
11	1.419021	74.53.140.153	10.10.1.4	SMTP	72	S: 334 VXNlcm5hbWU6
12	1.419595	10.10.1.4	74.53.140.153	SMTP	84	C: User: Z3VycGFydGFwQHBhdHJpb3RzLmlu
13	1.761484	74.53.140.153	10.10.1.4	SMTP	72	S: 334 UGFzc3dvcmQ6
14	1.762058	10.10.1.4	74.53.140.153	SMTP	72	C: Pass: cHVuamFiQDEyMw==
15	2.121738	74.53.140.153	10.10.1.4	SMTP	84	S: 235 Authentication succeeded

▶ Frame 6: 235 bytes on wire (1880 bits), 235 bytes captured (1880 bits)
▶ Ethernet II, Src: Netgear_d9:81:60 (00:1f:33:d9:81:60), Dst: Cradlepo_3c:17:c2 (00:e0:1c:3c:17:c2)
▶ Internet Protocol Version 4, Src: 74.53.140.153, Dst: 10.10.1.4
▶ Transmission Control Protocol, Src Port: 25, Dst Port: 1470, Seq: 1, Ack: 1, Len: 181

▼ Simple Mail Transfer Protocol

▼ Response: 220-xc90.websitewelcome.com ESMTP Exim 4.69 #1 Mon, 05 Oct 2009 01:05:54 -0500 \r\n

Response code: <domain> **Service ready (220)**

Response parameter: xc90.websitewelcome.com ESMTP Exim 4.69 #1 Mon, 05 Oct 2009 01:05:54 -0500

Response parameter: We do not authorize the use of this system to transport unsolicited, and/or bulk e-mail.

indicates that

- the device is running an SMTP server and is ready to receive email messages.
- additional information about the server and a warning that the server does not authorize the use of its system to send unsolicited or bulk emails.

smtp.pcap						
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help						
smtp						
No.	Time	Source	Destination	Protocol	Length	Info
6	0.727603	74.53.140.153	10.10.1.4	SMTP	235	S: 220-xc90.websitewelcome.com ESMTP Exim 4.69 #1 Mon, 05 Oct 2009 01:05:54 -0500 We do not authorize the use of th
7	0.732749	10.10.1.4	74.53.140.153	SMTP	63	C: EHLO GP
9	1.074123	74.53.140.153	10.10.1.4	SMTP	191	S: 250-xc90.websitewelcome.com Hello GP [122.162.143.157] SIZE 52428800 PIPELINING AUTH PLAIN LOGIN STARTTLS
10	1.076669	10.10.1.4	74.53.140.153	SMTP	66	C: AUTH LOGIN
11	1.419021	74.53.140.153	10.10.1.4	SMTP	72	S: 334 VXNlcm5hbWU6
12	1.419595	10.10.1.4	74.53.140.153	SMTP	84	C: User: Z3VycGFydGFwQHhhdHJpb3RzLmlu
13	1.761484	74.53.140.153	10.10.1.4	SMTP	72	S: 334 UGFzc3dvcmQ6
14	1.762058	10.10.1.4	74.53.140.153	SMTP	72	C: Pass: cHVuamFiQDEyMw==
15	2.121738	74.53.140.153	10.10.1.4	SMTP	84	S: 235 Authentication succeeded
16	2.122354	10.10.1.4	74.53.140.153	SMTP	90	C: MAIL FROM: <gurpartap@patriots.in>
17	2.464705	74.53.140.153	10.10.1.4	SMTP	62	S: 250 OK
18	2.465190	10.10.1.4	74.53.140.153	SMTP	93	C: RCPT TO: <raj_deol2002in@yahoo.co.in>
19	2.827648	74.53.140.153	10.10.1.4	SMTP	68	S: 250 Accepted
20	2.828143	10.10.1.4	74.53.140.153	SMTP	60	C: DATA
21	3.169619	74.53.140.153	10.10.1.4	SMTP	110	S: 354 Enter message, ending with "." on a line by itself
22	3.200683	10.10.1.4	74.53.140.153	SMTP	1514	C: DATA fragment, 1460 bytes
23	3.200726	10.10.1.4	74.53.140.153	SMTP	1514	C: DATA fragment, 1460 bytes
24	3.200744	10.10.1.4	74.53.140.153	SMTP	1514	C: DATA fragment, 1460 bytes
25	3.200763	10.10.1.4	74.53.140.153	SMTP	1514	[TCP Window Full] C: DATA fragment, 1460 bytes
26	3.203055	192.168.1.1	10.10.1.4	ICMP	590	Destination unreachable (Fragmentation needed)
28	3.203563	192.168.1.1	10.10.1.4	ICMP	590	Destination unreachable (Fragmentation needed)
29	3.204188	192.168.1.1	10.10.1.4	ICMP	590	Destination unreachable (Fragmentation needed)
30	3.204574	192.168.1.1	10.10.1.4	ICMP	590	Destination unreachable (Fragmentation needed)
38	4.002121	10.10.1.4	74.53.140.153	SMTP	1506	C: DATA fragment, 1452 bytes
39	4.002139	10.10.1.4	74.53.140.153	SMTP	1506	C: DATA fragment, 1452 bytes
41	4.342568	10.10.1.4	74.53.140.153	SMTP	1506	C: DATA fragment, 1452 bytes
42	4.342595	10.10.1.4	74.53.140.153	SMTP	1506	C: DATA fragment, 1452 bytes
44	4.366256	10.10.1.4	74.53.140.153	SMTP	1506	C: DATA fragment, 1452 bytes
45	4.366274	10.10.1.4	74.53.140.153	SMTP/I...	83	from: "Gurpartap Singh" <gurpartap@patriots.in>, subject: SMTP, (text/plain) (text/html) (text/plain) (text/plain)
52	4.756729	74.53.140.153	10.10.1.4	SMTP	82	S: 250 OK id=1Mugho-0003Dg-Un
54	7.271765	10.10.1.4	74.53.140.153	SMTP	60	C: QUIT
56	7.613407	74.53.140.153	10.10.1.4	SMTP	102	S: 221 xc90.websitewelcome.com closing connection

Username and password are
base64 encoded



What is Base64 encoding

- Base64 encoding is a way to represent binary data in a text format.
- It is often used in email messages, HTML pages, and other types of text-based communication to transfer binary data such as images, audio files, or other types of non-text data.

Index	Binary	Char	Index	Binary	Char	Index	Binary	Char	Index	Binary	Char
0	000000	A	16	010000	Q	32	100000	g	48	110000	w
1	000001	B	17	010001	R	33	100001	h	49	110001	x
2	000010	C	18	010010	S	34	100010	i	50	110010	y
3	000011	D	19	010011	T	35	100011	j	51	110011	z
4	000100	E	20	010100	U	36	100100	k	52	110100	0
5	000101	F	21	010101	V	37	100101	l	53	110101	1
6	000110	G	22	010110	W	38	100110	m	54	110110	2
7	000111	H	23	010111	X	39	100111	n	55	110111	3
8	001000	I	24	011000	Y	40	101000	o	56	111000	4
9	001001	J	25	011001	Z	41	101001	p	57	111001	5
10	001010	K	26	011010	a	42	101010	q	58	111010	6
11	001011	L	27	011011	b	43	101011	r	59	111011	7
12	001100	M	28	011100	c	44	101100	s	60	111100	8
13	001101	N	29	011101	d	45	101101	t	61	111101	9
14	001110	O	30	011110	e	46	101110	u	62	111110	+
15	001111	P	31	011111	f	47	101111	v	63	111111	/

Conversion

Source	Text (ASCII)	M								a								n							
	Octets	77 (0x4d)								97 (0x61)								110 (0x6e)							
Bits		0	1	0	0	1	1	0	1	0	1	1	0	0	0	0	1	0	1	1	0	1	1	1	0
Base64 encoded	Sextets	19								22								5							
	Character	T								W								F							
	Octets	84 (0x54)								87 (0x57)								70 (0x46)							

Source	Text (ASCII)	M								a															
	Octets	77 (0x4d)								97 (0x61)															
Bits		0	1	0	0	1	1	0	1	0	1	1	0	0	0	0	1	0	0						
Base64 encoded	Sextets	19								22								4							
	Character	T								W								E							
	Octets	84 (0x54)								87 (0x57)								69 (0x45)							

Source	Text (ASCII)	M																							
	Octets	77 (0x4d)																							
Bits		0	1	0	0	1	1	0	1	0	0	0	0												
Base64 encoded	Sextets	19								16				Padding				Padding							
	Character	T								Q				=				=							
	Octets	84 (0x54)								81 (0x51)				61 (0x3D)				61 (0x3D)							

Decoding Base64 to text

```
1 import base64
2
3 # Base64-encoded text
4 base64_text = "TWFu"
5
6 # Decode Base64-encoded text to ASCII
7 ascii_text = base64.b64decode(base64_text).decode('utf-8')
8
9 # Print the decoded ASCII text
10 print(ascii_text)
11
```

[8] ✓ 0.0s

...

Man



Decoding username

```
1 import base64
2
3 # Base64-encoded text
4 base64_text = "Z3VycGFydGFwQHBhdHJpb3RzLm1u"
5
6 # Decode Base64-encoded text to ASCII
7 ascii_text = base64.b64decode(base64_text).decode('utf-8')
8
9 # Print the decoded ASCII text
10 print(ascii_text)
11
```

[4] ✓ 0.0s

... gurpartap@patriots.in



1. right click

2

3

4

Protocol Preferences

Simple Mail Transfer Protocol

Decode Base64 encoded AUTH parameters

SMTP

66 C: AUTH LOGIN

72 S: 334 VXNlcm5hbWU6

84 C: User: Z3VycGFydGFwQHhhdHJp

72 S: 334 UGFzc3dvcmQ6

72 C: Pass: chVuamFiQDEyMw==

84 S: 235 Authentication succeeded

90 C: MAIL FROM: <gurpartap@patriots.in>

62 S: 250 OK

93 C: RCPT TO: <raj_deol2002in@patriots.in>

68 S: 250 Accepted

60 C: DATA

110 S: 354 Enter message, ending with a dot on a line by itself

1514 C: DATA fragment, 1460 bytes

1514 C: DATA fragment, 1460 bytes

1514 C: DATA fragment, 1460 bytes

1514 [TCP Window Full] C: DATA fragment, 1460 bytes

590 Destination unreachable (Fragmentation needed) (Seq=1514, Win=0, Len=1460)

590 Destination unreachable (Fragmentation needed) (Seq=1514, Win=0, Len=1460)

590 Destination unreachable (Fragmentation needed) (Seq=1514, Win=0, Len=1460)

590 Destination unreachable (Fragmentation needed) (Seq=1514, Win=0, Len=1460)

1506 C: DATA fragment, 1452 bytes

1506 C: DATA fragment, 1452 bytes

1506 C: DATA fragment, 1452 bytes

1506 C: DATA fragment, 1452 bytes

1506 C: DATA fragment, 1452 bytes

83 from: "Gurpartap Singh" <gurpartap@patriots.in> subject: SMTP (4096 bytes)

Open Simple Mail Transfer Protocol preferences...

☒ Reassemble SMTP command and response lines spanning multiple TCP segments

☒ Reassemble SMTP DATA commands spanning multiple TCP segments

☐ Decode Base64 encoded AUTH parameters

Frame

Ethernet

Internet Protocol Version 4

Transmission Control Protocol

Simple Mail Transfer Protocol

[No Title]

captured (672 bits)

TCP port(s): 25...



smtp						
No.	Time	Source	Destination	Protocol	Length	Info
6	0.727603	74.53.140.153	10.10.1.4	SMTP	235	S: 220-xc90.websitewelcome.com ESMTP
7	0.732749	10.10.1.4	74.53.140.153	SMTP	63	C: EHLO GP
9	1.074123	74.53.140.153	10.10.1.4	SMTP	191	S: 250-xc90.websitewelcome.com Hello
10	1.076669	10.10.1.4	74.53.140.153	SMTP	66	C: AUTH LOGIN
11	1.419021	74.53.140.153	10.10.1.4	SMTP	72	S: 334 Username:
12	1.419595	10.10.1.4	74.53.140.153	SMTP	84	C: User: gurpartap@patriots.in
13	1.761484	74.53.140.153	10.10.1.4	SMTP	72	S: 334 Password:
14	1.762058	10.10.1.4	74.53.140.153	SMTP	72	C: Pass: punjab@123
15	2.121738	74.53.140.153	10.10.1.4	SMTP	84	S: 235 Authentication succeeded
16	2.122354	10.10.1.4	74.53.140.153	SMTP	90	C: MAIL FROM: <gurpartap@patriots.in
17	2.464705	74.53.140.153	10.10.1.4	SMTP	62	S: 250 OK
18	2.465190	10.10.1.4	74.53.140.153	SMTP	93	C: RCPT TO: <raj_deol2002in@yahoo.co
19	2.827648	74.53.140.153	10.10.1.4	SMTP	68	S: 250 Accepted

Frame 10: 66 bytes on wire (528 bits), 66 bytes captured (528 bits)
 Ethernet II, Src: Cradlepo_3c:17:c2 (00:e0:1c:3c:17:c2), Dst: Netgear_d9:81:60 (00:1f:33:d9:81:60)
 Internet Protocol Version 4, Src: 10.10.1.4, Dst: 74.53.140.153
 Transmission Control Protocol, Src Port: 1470, Dst Port: 25, Seq: 10, Ack: 319, Len: 12
 Simple Mail Transfer Protocol
 Command Line: AUTH LOGIN\r\n
 Command: AUTH
 Request parameter: LOGIN

smtp						
No.	Time	Source	Destination	Protocol	Length	Info
6	0.727603	74.53.140.153	10.10.1.4	SMTP	235	S: 220-xc90.websitewelo
7	0.732749	10.10.1.4	74.53.140.153	SMTP	63	C: EHLO GP
9	1.074123	74.53.140.153	10.10.1.4	SMTP	191	S: 250-xc90.websitewelo
10	1.076669	10.10.1.4	74.53.140.153	SMTP	66	C: AUTH LOGIN
11	1.419021	74.53.140.153	10.10.1.4	SMTP	72	S: 334 Username:
12	1.419595	10.10.1.4	74.53.140.153	SMTP	84	C: User: gurpartap@patr
13	1.761484	74.53.140.153	10.10.1.4	SMTP	72	S: 334 Password:
14	1.762058	10.10.1.4	74.53.140.153	SMTP	72	C: Pass: punjab@123
15	2.121738	74.53.140.153	10.10.1.4	SMTP	84	S: 235 Authentication s
16	2.122354	10.10.1.4	74.53.140.153	SMTP	90	C: MAIL FROM: <gurparta
17	2.464705	74.53.140.153	10.10.1.4	SMTP	62	S: 250 OK
18	2.465190	10.10.1.4	74.53.140.153	SMTP	93	C: RCPT TO: <raj_deol20
19	2.827648	74.53.140.153	10.10.1.4	SMTP	68	S: 250 Accepted

▶ Frame 11: 72 bytes on wire (576 bits), 72 bytes captured (576 bits)
 ▶ Ethernet II, Src: Netgear_d9:81:60 (00:1f:33:d9:81:60), Dst: Cradlepo_3c:17:c2 (00:e0:1c:3c:17:c2)
 ▶ Internet Protocol Version 4, Src: 74.53.140.153, Dst: 10.10.1.4
 ▶ Transmission Control Protocol, Src Port: 25, Dst Port: 1470, Seq: 319, Ack: 22, Len: 18
 ▶ Simple Mail Transfer Protocol
 ▶ Response: 334 VXNlcm5hbWU6\r\n
 Response code: AUTH input (334)
 Response parameter: Username:

smtp					
No.	Time	Source	Destination	Protocol	Length Info
6	0.727603	74.53.140.153	10.10.1.4	SMTP	235 S: 220-xc90.websitewelcome.com ESM
7	0.732749	10.10.1.4	74.53.140.153	SMTP	63 C: EHLO GP
9	1.074123	74.53.140.153	10.10.1.4	SMTP	191 S: 250-xc90.websitewelcome.com Hel
10	1.076669	10.10.1.4	74.53.140.153	SMTP	66 C: AUTH LOGIN
11	1.419021	74.53.140.153	10.10.1.4	SMTP	72 S: 334 Username:
12	1.419595	10.10.1.4	74.53.140.153	SMTP	84 C: User: gurpartap@patriots.in
13	1.761484	74.53.140.153	10.10.1.4	SMTP	72 S: 334 Password:
14	1.762058	10.10.1.4	74.53.140.153	SMTP	72 C: Pass: punjab@123
15	2.121738	74.53.140.153	10.10.1.4	SMTP	84 S: 235 Authentication succeeded
16	2.122354	10.10.1.4	74.53.140.153	SMTP	90 C: MAIL FROM: <gurpartap@patriots.
17	2.464705	74.53.140.153	10.10.1.4	SMTP	62 S: 250 OK
18	2.465190	10.10.1.4	74.53.140.153	SMTP	93 C: RCPT TO: <raj_deol2002in@yahoo.
19	2.827648	74.53.140.153	10.10.1.4	SMTP	68 S: 250 Accepted
- Frame 12: 84 bytes on wire (672 bits), 84 bytes captured (672 bits) - Ethernet II, Src: Cradlepo_3c:17:c2 (00:e0:1c:3c:17:c2), Dst: Netgear_d9:81:60 (00:1f:33:d9:81:60) - Internet Protocol Version 4, Src: 10.10.1.4, Dst: 74.53.140.153 - Transmission Control Protocol, Src Port: 1470, Dst Port: 25, Seq: 22, Ack: 337, Len: 30 - Simple Mail Transfer Protocol - Username: gurpartap@patriots.in					

Time	Source	Destination	Protocol	Length	Info
6 0.727603	74.53.140.153	10.10.1.4	SMTP	235	S: 220-xc90.websitewelcome.com ESMTP Exim 4.69 #1 Mon, 05 Oct 200
7 0.732749	10.10.1.4	74.53.140.153	SMTP	63	C: EHLO GP
9 1.074123	74.53.140.153	10.10.1.4	SMTP	191	S: 250-xc90.websitewelcome.com Hello GP [122.162.143.157] SIZE
10 1.076669	10.10.1.4	74.53.140.153	SMTP	66	C: AUTH LOGIN
11 1.419021	74.53.140.153	10.10.1.4	SMTP	72	S: 334 Username:
12 1.419595	10.10.1.4	74.53.140.153	SMTP	84	C: User: gurpartap@patriots.in
13 1.761484	74.53.140.153	10.10.1.4	SMTP	72	S: 334 Password:
14 1.762058	10.10.1.4	74.53.140.153	SMTP	72	C: Pass: punjab@123
15 2.121738	74.53.140.153	10.10.1.4	SMTP	84	S: 235 Authentication succeeded
16 2.122354	10.10.1.4	74.53.140.153	SMTP	90	C: MAIL FROM: <gurpartap@patriots.in>
17 2.464705	74.53.140.153	10.10.1.4	SMTP	62	S: 250 OK
18 2.465190	10.10.1.4	74.53.140.153	SMTP	93	C: RCPT TO: <raj_deol2002in@yahoo.co.in>
19 2.827648	74.53.140.153	10.10.1.4	SMTP	68	S: 250 Accepted
20 2.828143	10.10.1.4	74.53.140.153	SMTP	60	C: DATA
21 3.169619	74.53.140.153	10.10.1.4	SMTP	110	S: 354 Enter message, ending with "." on a line by itself
22 3.200683	10.10.1.4	74.53.140.153	SMTP	1514	C: DATA fragment, 1460 bytes
23 3.200726	10.10.1.4	74.53.140.153	SMTP	1514	C: DATA fragment, 1460 bytes
24 3.200744	10.10.1.4	74.53.140.153	SMTP	1514	C: DATA fragment, 1460 bytes
25 3.200763	10.10.1.4	74.53.140.153	SMTP	1514	[TCP Window Full] C: DATA fragment, 1460 bytes
26 3.203055	192.168.1.1	10.10.1.4	ICMP	590	Destination unreachable (Fragmentation needed)
28 3.203563	192.168.1.1	10.10.1.4	ICMP	590	Destination unreachable (Fragmentation needed)
29 3.204188	192.168.1.1	10.10.1.4	ICMP	590	Destination unreachable (Fragmentation needed)
30 3.204574	192.168.1.1	10.10.1.4	ICMP	590	Destination unreachable (Fragmentation needed)
38 4.002121	10.10.1.4	74.53.140.153	SMTP	1506	C: DATA fragment, 1452 bytes
39 4.002139	10.10.1.4	74.53.140.153	SMTP	1506	C: DATA fragment, 1452 bytes
41 4.342568	10.10.1.4	74.53.140.153	SMTP	1506	C: DATA fragment, 1452 bytes
42 4.342595	10.10.1.4	74.53.140.153	SMTP	1506	C: DATA fragment, 1452 bytes
44 4.366256	10.10.1.4	74.53.140.153	SMTP	1506	C: DATA fragment, 1452 bytes
45 4.366274	10.10.1.4	74.53.140.153	SMTP/I...	83	from: "Gurpartap Singh" <gurpartap@patriots.in>, subject: SMTP,

ignore them



Show email content (14 fragments, Wireshark reassemble them at 45)

No.	Time	Source	Destination	Protocol	Length	Info
40	4.342535	74.53.140.153	10.10.1.4	TCP	60	25 → 1470 [ACK] Seq=463 Ack=5959 Win=17424 Len=0
41	4.342568	10.10.1.4	74.53.140.153	SMTP	1506	C: DATA fragment, 1452 bytes
42	4.342595	10.10.1.4	74.53.140.153	SMTP	1506	C: DATA fragment, 1452 bytes
43	4.366241	74.53.140.153	10.10.1.4	TCP	60	[TCP ACKed unseen segment] 25 → 1470 [ACK] Seq=463 Ack=7411 Win=20328 Len=0
44	4.366256	10.10.1.4	74.53.140.153	SMTP	1506	C: DATA fragment, 1452 bytes
45	4.366274	10.10.1.4	74.53.140.153	SMTP/IMF	83	from: "Gurpartap Singh" <gurpartap@patriots.in>, subject: SMTP, (text/plain) (text/html) (text/plain)
46	4.389163	74.53.140.153	10.10.1.4	TCP	60	25 → 1470 [ACK] Seq=463 Ack=8863 Win=23232 Len=0
47	4.413523	74.53.140.153	10.10.1.4	TCP	60	25 → 1470 [ACK] Seq=463 Ack=10315 Win=26136 Len=0
48	4.708119	74.53.140.153	10.10.1.4	TCP	60	25 → 1470 [ACK] Seq=463 Ack=11767 Win=29040 Len=0
49	4.730686	74.53.140.153	10.10.1.4	TCP	60	25 → 1470 [ACK] Seq=463 Ack=13219 Win=31944 Len=0
50	4.754784	74.53.140.153	10.10.1.4	TCP	60	25 → 1470 [ACK] Seq=463 Ack=14671 Win=34848 Len=0
51	4.756231	74.53.140.153	10.10.1.4	TCP	60	25 → 1470 [ACK] Seq=463 Ack=14700 Win=34848 Len=0
52	4.756729	74.53.140.153	10.10.1.4	SMTP	82	S: 250 OK id=1Mugho-0003Dg-Un
53	4.895535	10.10.1.4	74.53.140.153	TCP	54	1470 → 25 [ACK] Seq=14700 Ack=491 Win=65045 Len=0

- ▶ Frame 45: 83 bytes on wire (664 bits), 83 bytes captured (664 bits)
- ▶ Ethernet II, Src: Cradlepo_3c:17:c2 (00:e0:1c:3c:17:c2), Dst: Netgear_d9:81:60 (00:1f:33:d9:81:60)
- ▶ Internet Protocol Version 4, Src: 10.10.1.4, Dst: 74.53.140.153
- ▶ Transmission Control Protocol, Src Port: 1470, Dst Port: 25, Seq: 14671, Ack: 463, Len: 29
- ▼ Simple Mail Transfer Protocol

Email consists of 14 fragments

- C: .
- ▶ [14 DATA fragments (15156 bytes): #22(1460), #23(1460), #24(1460), #25(1460), #26(508), #28(508), #29(508), #30(508), #38(1452), #39(1452), #41(1452), #42(1452), #44(1452), #45(24)]
- ▼ Internet Message Format
- ▶ From: "Gurpartap Singh" <gurpartap@patriots.in>, 1 item
- ▶ To: <raj_deol2002in@yahoo.co.in>, 1 item
- Subject: SMTP
- Date: Mon, 5 Oct 2009 11:36:07 +0530
- Message-ID: <000301ca4581\$ef9e57f0\$cedb07d0\$@in>
- MIME-Version: 1.0
- ▶ Content-Type: multipart/mixed;\r\n\tboundary="-----_NextPart_000_0004_01CA45B0.095693F0"
- X-Mailer: Microsoft Office Outlook 12.0
- Thread-Index: AcpFgem9BvjZEDeR1Kh8i+hUyVo0A==
- Content-Language: en-us
- ▶ Unknown-Extension [truncated]: x-cr-hashedpuzzle: SeA= AAR2 ADaH BpiO C4G1 D1gW FNB1 FPKr Fn+W HFcp HnYJ JO7s Kum6 KytW LFcI LjUt;1;cgBhAGoAXwBkAGUAbwBsADIAMAawADIAaQBuaEAAeQBhAGGAbwB
- ▶ Unknown-Extension: x-cr-puzzleid: {CAA37F59-1850-45C7-8540-AA27696B5398} (Contact Wireshark developers if you want this supported.)
- ▶ MIME Multipart Media Encapsulation, Type: multipart/mixed, Boundary: "-----_NextPart_000_0004_01CA45B0.095693F0"

The whole email

email client: office outlook 12.0



Show email content (text content)

```
Content-Language: en-us
▶ Unknown-Extension [truncated]: x-cr-hashedpuzzle: SeA= AAR2 ADaH BpiO C4G1 D1gW FNB1 FPKR Fn+W HFCP HnYJ J07s Kum6 KytW LFcI LjUt;1;cgB
▶ Unknown-Extension: x-cr-puzzleid: {CAA37F59-1850-45C7-8540-AA27696B5398} (Contact Wireshark developers if you want this supported.)
▼ MIME Multipart Media Encapsulation, Type: multipart/mixed, Boundary: "-----_NextPart_000_0004_01CA45B0.095693F0"
  [Type: multipart/mixed]
  Preamble: 546869732069732061206d756c746970617274206d65737361676520696e204d494d4520...
  First boundary: -----_NextPart_000_0004_01CA45B0.095693F0\r\n
  ▼ Encapsulated multipart part: (multipart/alternative)
    Content-Type: multipart/alternative;\r\n\tboundary="-----_NextPart_001_0005_01CA45B0.095693F0"\r\n\r\n
    ▼ MIME Multipart Media Encapsulation, Type: multipart/alternative, Boundary: "-----_NextPart_001_0005_01CA45B0.095693F0"
      [Type: multipart/alternative]
      Preamble: 0d0a
      First boundary: -----_NextPart_001_0005_01CA45B0.095693F0\r\n
      ▼ Encapsulated multipart part: (text/plain)
        Content-Type: text/plain;\r\n\tcharset="us-ascii"\r\n
        Content-Transfer-Encoding: 7bit\r\n\r\n
        ▼ Line-based text data: text/plain (12 lines)
          Hello\r\n
          \r\n
          \r\n
          \r\n
          I send u smtp pcap file \r\n
          \r\n
          Find the attachment\r\n
          \r\n
          \r\n
          \r\n
          GPS\r\n
          \r\n
          Boundary: \r\n-----_NextPart_001_0005_01CA45B0.095693F0\r\n
        ▶ Encapsulated multipart part: (text/html)
          Last boundary: \r\n-----_NextPart_001_0005_01CA45B0.095693F0--\r\n
```

contain multiple file types

first part



```

Internet Message Format
  ▶ From: "Gurpartap Singh" <gurpartap@patriots.in>, 1 item
  ▶ To: <raj_deol2002in@yahoo.co.in>, 1 item
    Subject: SMTP
    Date: Mon, 5 Oct 2009 11:36:07 +0530
    Message-ID: <000301ca4581$ef9e57f0$cedb07d0$@in>
    MIME-Version: 1.0
  ▶ Content-Type: multipart/mixed;\r\n\tboundary="-----_NextPart_000_0004_01CA45B0.095693F0"
    X-Mailer: Microsoft Office Outlook 12.0
    Thread-Index: AcpFgem9BvjJZEDeR1Kh8i+hUyVo0A==
    Content-Language: en-us
  ▶ Unknown-Extension [truncated]: x-cr-hashedpuzzle: SeA= AAR2 ADaH Bpi0 C4G1 D1gW FNB1 FPKR Fn+W HFCP HnYJ JO7s Kum6 KytW LFcI LjUt;1;cgBhAGoAXwBkAGUA
  ▶ Unknown-Extension: x-cr-puzzleid: {CAA37F59-1850-45C7-8540-AA27696B5398} (Contact Wireshark developers if you want this supported.)
  ▼ MIME Multipart Media Encapsulation, Type: multipart/mixed, Boundary: "-----_NextPart_000_0004_01CA45B0.095693F0"
    [Type: multipart/mixed]
    Preamble: 546869732069732061206d756c746970617274206d65737361676520696e204d494d4520...
    First boundary: -----= NextPart 000 0004 01CA45B0.095693F0\r\n
    ▼ Encapsulated multipart part: (multipart/alternative)
      Content-Type: multipart/alternative;\r\n\tboundary="-----_NextPart_001_0005_01CA45B0.095693F0"\r\n\r\n
      ▶ MIME Multipart Media Encapsulation, Type: multipart/alternative, Boundary: "-----_NextPart_001_0005_01CA45B0.095693F0"
      Boundary: \r\n-----= NextPart 000 0004 01CA45B0.095693F0\r\n
    ▼ Encapsulated multipart part: (text/plain)
      Content-Type: text/plain;\r\n\tname="NEWS.txt"\r\n
      Content-Transfer-Encoding: quoted-printable\r\n
      Content-Disposition: attachment;\r\n\tfilename="NEWS.txt"\r\n\r\n
      ▶ Line-based text data: text/plain (114 lines)
      Boundary: \r\n-----= NextPart 000 0004 01CA45B0.095693F0\r\n
    ▶ Encapsulated multipart part: (text/plain)
      Last boundary: \r\n-----= NextPart_000_0004_01CA45B0.095693F0--\r\n

```

part 1

part 2

part 3



Show email content (Alternative approach: follow TCP)

smtp

No.	Time	Source	Destination	Protocol	Length	Info
6	0.727603	74.53.140.153	10.10.1.4	SMTP	235	S: 220-xc90.websitewelcome.com ESMTP Exim 4.69 #1 Mon, 05 Oct 2009 01:05:54 -0500 We do not authorize the
7	0.732749	10.10.1.4	74.53.140.153	SMTP	63	C: EHLO GP
9	1.074123	74.53.140.153	10.10.1.4	SMTP	191	S: 250-xc90.websitewelcome.com Hello GP [122.162.143.157] SIZE 52428800 PIPELINING AUTH PLAIN LOGIN
10	1.076669	10.10.1.4	74.53.140.153	SMTP	66	C: AUTH LOGIN
11	1.419021	74.53.140.153	10.10.1.4	SMTP	72	S: 334 Username:
12	1.419595	10.10.1.4	74.53.140.153	SMTP	84	C: User: gurpartap@patriots.in
13	1.761484	74.53.140.153	10.10.1.4	SMTP	72	S: 334 Password:
14	1.762058	10.10.1.4	74.53.140.153	SMTP	72	C: Pass: punjab@123
15	2.121738	74.53.140.153	10.10.1.4	SMTP	84	S: 235 Authentication succeeded
16	2.122354	10.10.1.4	74.53.140.153	SMTP	90	C: MAIL FROM: <gurpartap@patriots.in>
17	2.464705	74.53.140.153	10.10.1.4	SMTP	62	S: 250 OK
18	2.465190	10.10.1.4	74.53.140.153	SMTP	93	C: RCPT TO: <raj_deol2002in@yahoo.co.in>
19	2.827648	74.53.140.153	10.10.1.4	SMTP	68	S: 250 Accepted
20	2.828143	10.10.1.4	74.53.140.153	SMTP	60	C: DATA
21	3.169619	74.53.140.153	10.10.1.4	SMTP	110	S: 354 Enter message, ending with
22	3.200683	10.10.1.4	74.53.140.153	SMTP	1514	C: DATA fragment, 1460 bytes
23	3.200726	10.10.1.4	74.53.140.153	SMTP	1514	C: DATA fragment, 1460 bytes
24	3.200744	10.10.1.4	74.53.140.153	SMTP	1514	C: DATA fragment, 1460 bytes
25	3.200763	10.10.1.4	74.53.140.153	SMTP	1514	[TCP Window Full] C: DATA fragment,
26	3.203055	192.168.1.1	10.10.1.4	ICMP	590	Destination unreachable (Fragmenta
28	3.203562	192.168.1.1	10.10.1.4	ICMP	590	Destination unreachable (Fragmenta

▶ Frame 20: 60 bytes on wire (480 bits), 60 bytes captured (480 bits)

▶ Ethernet II, Src: Cradlepo_3c:17:c2 (00:e0:1c:3c:17:c2), Dst: Netgear_d9:81:60 (00:1f:33:d9:81:60)

▶ Internet Protocol Version 4, Src: 10.10.1.4, Dst: 74.53.140.153

▶ Transmission Control Protocol, Src Port: 1470, Dst Port: 25, Seq: 145, Ack: 407, Len: 6

▼ Simple Mail Transfer Protocol

Command Line: DATA\r\nCommand: DATA

1. Right click

2

3

Mark/Unmark PacketCtrl+M

Ignore/Unignore PacketCtrl+D

Set/Unset Time ReferenceCtrl+T

Time Shift...Ctrl+Shift+T

Packet Comment...Ctrl+Alt+C

Edit Resolved Name

Apply as Filter

Prepare as Filter

Conversation Filter

Colorize Conversation

SCTP

FollowTCP StreamCtrl+Alt+Shift+T

CopyUDP StreamCtrl+Alt+Shift+U

Protocol PreferencesTLS StreamCtrl+Alt+Shift+S

Decode As...HTTP StreamCtrl+Alt+Shift+H

Show Packet in New WindowHTTP/2 Stream

QUIC Stream



```

220-xc90.websitewelcome.com ESMTP Exim 4.69 #1 Mon, 05 Oct 2009 01:05:54 -0500
220-We do not authorize the use of this system to transport unsolicited,
220 and/or bulk e-mail.
EHLO GP
250-xc90.websitewelcome.com Hello GP [122.162.143.157]
250-SIZE 52428800
250-PIPELINING
250-AUTH PLAIN LOGIN
250-STARTTLS
250 HELP
AUTH LOGIN
334 VXNlcm5hbWU6
Z3VycGFydGFwQHBhdHJpb3RzLmlu
334 UGFzc3dvcmQ0
cHVuamFiQDEyMw==
235 Authentication succeeded
MAIL FROM: <gurpartap@patriots.in>
250 OK
RCPT TO: <raj_deol2002in@yahoo.co.in>
250 Accepted
DATA
354 Enter message, ending with "." on a line by itself
From: "Gurpartap Singh" <gurpartap@patriots.in>
To: <raj_deol2002in@yahoo.co.in>
Subject: SMTP
Date: Mon, 5 Oct 2009 11:36:07 +0530
Message-ID: <000301ca4581$ef9e57f0$cedb07d0$@in>
MIME-Version: 1.0
Content-Type: multipart/mixed;
        boundary="-----_NextPart_000_0004_01CA45B0.095693F0"
X-Mailer: Microsoft Office Outlook 12.0
Thread-Index: AcpFgem9BvjjZEDeR1Kh8i+hUyVo0A==
Content-Language: en-us
x-cr-hashedpuzzle: SeA= AAR2 ADaH Bpi0 C4G1 D1gW FNB1 FPKR Fn+W HFCP HnYJ J07s Kum6 KytW LFcI LjUt;
1;cgBhAGoAXwBkAGUAbwBsADIAMAAwADIAaQBUEAAeQBhAGgAbwBvAC4AYwBvAC4AaQBUEAA==;Sosha1_v1;7;{CAA37F59-1850-45C7-8540-
AA27696B5398};ZwB1AHIAcABhAHIAcABhAHAAQABwAGEAdABYAGkAbwB0AHMALgBpAG4A;Mon, 05 Oct 2009 06:06:01 GMT;UwBNAFQAUA==
x-cr-puzzleid: {CAA37F59-1850-45C7-8540-AA27696B5398}

This is a multipart message in MIME format.

-----_NextPart_000_0004_01CA45B0.095693F0
Content-Type: multipart/alternative;
        boundary="-----_NextPart_001_0005_01CA45B0.095693F0"

-----_NextPart_001_0005_01CA45B0.095693F0
Content-Type: text/plain;
        charset="us-ascii"
Content-Transfer-Encoding: 7bit
Hello

I send u smtp pcap file

Find the attachment

GPS

```

commands

Email message



Questions you need to answer

- When did it happen?
 - the timestamps email
- The email client
 - outlook
- The content of the email
 - multiple parts
- What was the approach?
 - ports (sender and receiver)
 - IP addresses (sender and receiver)
 - Mac addresses (sender and receiver)



Extra credits

- Sniff SMTP traffic

